Quantum minimum-cut query lower bounds for weighted graph input models

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Editorial note. Faithful editable selection of author arXiv:2011.09823v3 (2021-05-24), CC BY-NC-SA4.0, for noncommercial research with attribution and share-alike retained. The complete independent Section5 lower-bound family, Theorems35,37,40, is grouped as one item across query models and edge-weight regimes, retaining exact integer/divisibility/error promises, both cut-weight and cut-shore tasks, graph constructions, oracle simulations, local claims and proofs. Exact credited connectivity/counting facts33–34, graph definitions and quantum input-oracle models are retained. Original adjacency-gadget PDF and full bibliography retained. Query/time-upper-bound references are presented explicitly by original Theorem21/Section4 numbers; counters resume their original numbers. Original author mathematics is not corrected or supplemented. No upper-algorithm technical by-product is counted as a separate result.

Allowed standard prerequisites (editorial).

- Connectivity query lower bounds. Exact original Theorem33 [DHHM06] retained as prior context.
- Approximate-degree counting bound and quantum query composition. Exact original Theorem34 [NW99] and composition citations [HLS07,Rei11a] retained, without recursively reproducing prior literature.

2 Preliminaries

For a natural number $n \geq 1$ we let $[n] = \{1, \ldots, n\}$. For a real number x we let $\lfloor x \rceil$ denote the closest integer to x.

2.1 Graph basics and notation

Let V be a finite set and $V^{(2)}$ the set of all subsets of V of cardinality 2. We represent a weighted undirected graph as a pair $G = (V, w)$ where $w : V^{(2)} \to \mathbb{R}$ is a non-negative function. We let $V(G)$ be the vertex set of a graph G and $E(G) = \{e \in V^{(2)} : w(e) > 0\}$ be the set of edges of G. We extend the weight function to sets $S \subseteq V^{(2)}$ by $w(S) = \sum_{e \in S} w(e)$. We say that G is *simple* if $w : V^{(2)} \to \{0, 1\}$ and in this case also denote G as $G = (V, E)$, where E is the set of edges. We call the ratio of the largest edge weight of G to the smallest the *edge-weight ratio* of G.

For a subset $X \subseteq V$ we use the shorthand $\overline{X} = V \setminus X$, and we say X is *non-trivial* if $\emptyset \neq X \subsetneq V$. For disjoint sets $X, Y \subseteq V$ we use $E(X, Y)$ for the set of edges with one endpoint in X and one endpoint in Y. For a non-trivial set X, let $\Delta_G(X) = \{\{i, j\} \in E(G) : i \in X, j \in \overline{X}\}$ be the set of edges of G with one endpoint in X and one endpoint in $\overline{X}$. A *cut* of G is a set of the form $\Delta_G(X)$ for some non-trivial set X. We call X and $\overline{X}$ the *shores* of the cut $\Delta_G(X)$. We call a cut of the form $\Delta_G(\{u\})$ a *star* cut, and refer to all other cuts as *non-star* cuts. The *weight* of a cut S is $w(S)$, which in the case of a simple graph equals $|S|$. We let $\lambda(G) = \min_{\emptyset \neq X \subsetneq V} w(\Delta_G(X))$ be the minimum weight of a cut in G. We call a cut realizing this bound a *minimum cut*. We call a cut $\Delta_G(X)$ satisfying $w(\Delta_G(X)) \leq \alpha\lambda(G)$ an α-near minimum cut. In the case where G is simple we call $\lambda(G)$ the *edge connectivity* of G. We will only use the term edge connectivity in the context of unweighted graphs.

2.3 Quantum query and computational models

For general background on the quantum query model we refer the reader to [HLŠ07]. Here we restrict ourselves to describing the quantum implementation of the input oracles in the adjacency matrix and adjacency array models.

In the adjacency matrix model, on input a weighted graph $G = (V, w)$, classically one can query any $\{u, v\} \in V^{(2)}$ and receive the answer $w(\{u, v\})$. We now describe how to model this by a quantum query. We will assume that the edge weights are given as binary decimal numbers with M_1 bits before the decimal and M_2 bits after the decimal for a total of $M = M_1 + M_2$ bits. The state of the quantum query algorithm will have three registers, a query register, an answer register, and a workspace register. The state of the algorithm will in general be in a superposition of the basis states $|\{u, v\}\rangle|b\rangle|a\rangle$ where $\{u, v\} \in V^{(2)}, b \in \{0, 1\}^M$ and $a \in \mathcal{A}$ for an arbitrary finite set $\mathcal{A}$. On input graph $G = (V, w)$, the input oracle O_G acts on a basis state $|\{u, v\}\rangle|b\rangle|a\rangle$ as

$$\mathsf{O}_G|\{u, v\}\rangle|b\rangle|a\rangle = |\{u, v\}\rangle|b \oplus w(\{u, v\})\rangle|a\rangle \ .$$

In the adjacency array model, on input a weighted n-vertex graph $G = (V, w)$ one can make two types of queries. In the first type, one can query a vertex $v \in V$ and receive its degree $\deg(v)$. The second type is specified by a family of functions $\{f_v : [\deg(v)] \to V\}_{v \in V}$ such that $f_v(i)$ corresponds to the ith neighbor of vertex v (according to some arbitrary but fixed ordering). A query consists of a pair (v, i) for $i \in [\deg(v)]$ and the returned answer is the pair $(f_v(i), w(\{v, f_v(i)\}))$. In this paper we will only need to model the second type of query quantumly. This is because our upper bounds are larger than n so we can let the algorithm classically query all degrees at the start of the

algorithm, and in our lower bound on the query complexity of edge connectivity for weighted graphs we assume the algorithm already knows the degree of every vertex. The state of the quantum query algorithm will again have a query register, an answer register, and a workspace register, with the state of the algorithm in general being in a superposition of the basis states $|(v,i)\rangle|x\rangle|b\rangle|a\rangle$ where $v \in V, i \in [\deg(v)], x \in \{0,\ldots,n-1\}, b \in \{0,1\}^M$, and $a \in \mathcal{A}$ for an arbitrary finite set $\mathcal{A}$. We further let $\tau : V \to \{0,1,\ldots,n-1\}$ be a bijection where $|V| = n$. Then the input oracle O_G acts on a basis state in the following way:

$$\mathsf{O}_G|(v,i)\rangle|x\rangle|b\rangle|a\rangle = |(v,i)\rangle|x + \tau(f_v(i)) \bmod n\rangle|b \oplus w(\{v, f_v(i)\})\rangle|a\rangle \ .$$

In Section 4 of the source we will further show that our query algorithms can be implemented in a time efficient manner. We analyze the time complexity in terms of the standard quantum circuit model augmented with two types of oracles. One is the oracle for the input, either in the adjacency matrix or array model, and the second is an oracle to a classical memory of $\tilde{O}(n)$ bits. The latter corresponds to a *quantum random-access-memory* or *QRAM*. We further assume that we can classically update a value in this $\tilde{O}(n)$ bit classical memory in time $\tilde{O}(1)$. The assumption of QRAM access is also required for the time efficiency of the sparsifier construction in [AdW20] which our algorithms build on, and in fact is a necessary (but sometimes inexplicit) assumption in the time analysis of many quantum algorithms for graph problems, e.g. [DHHM06, AŠ06, BCJ+13].

5 Lower bounds

In this section we present lower bounds on the complexity of edge connectivity and weighted minimum cut.

First we describe some existing lower bounds for the case of simple graphs. Let CON_n be the problem of deciding if an input *simple* graph on n vertices is connected or not. This is a special case of edge connectivity, where one wants to decide if the edge connectivity is zero or positive. Dürr, Heiligman, Høyer and Mhalla [DHHM06] proved the following quantum query lower bounds on the complexity of CON_n.

Theorem 33 ([DHHM06])**.** *The bounded-error quantum query complexity of* CON_n *is* $\Theta(n^{3/2})$ *in the adjacency matrix model and* $\Theta(n)$ *in the adjacency array model.*

This theorem shows that, in the adjacency matrix model, Theorem 21 of the source is tight up to polylogarithmic factors for simple graphs. For the adjacency array model there is still a gap between the $\Omega(n)$ lower bound from Theorem 33 and the $\tilde{O}(\sqrt{mn})$ upper bound for simple graphs given by Theorem 21 of the source.

For the minimum cut problem in a weighted graph we prove separate and distinct lower bounds for the adjacency matrix model and the adjacency array model. All our lower bounds essentially follow by forcing the algorithm to solve a counting problem in order to compute the weight of a minimum cut. We then use the following theorem by Nayak and Wu that gives a lower bound on the quantum query complexity of exact counting.

Theorem 34 ([NW99, Corollary 1.2])**.** *Let $k, N \in \mathbb{N}$ with $2k+1 \leq N$. Assume query access to $x \in \{0,1\}^N$ with the promise that $|x| = k+1$ or $|x| = k-1$. Any quantum algorithm that correctly decides whether $|x| = k+1$ or $|x| = k-1$ with probability at least $2/3$ must make $\Omega(\sqrt{Nk})$ queries.*

5.1 Adjacency matrix model

In the adjacency matrix model we show that for any integer $1 \leq \tau \leq (\lfloor n/2 \rfloor - 1)/2$, in the worst case $\Omega(n^{3/2}\sqrt{\tau})$ adjacency matrix queries are needed to compute the weight of a minimum cut of a graph with edge weights in $\{1, \tau\}$. This matches the upper bound in Theorem 21 of the source, and hence settles the quantum query complexity of weighted minimum cut in the adjacency matrix model. For $\tau = 1$ this reproduces the aforementioned $\Omega(n^{3/2})$ bound which follows from [DHHM06].

Theorem 35. *Let $n, \tau \in \mathbb{N}$ satisfy $1 \leq \tau \leq (\lfloor n/2 \rfloor - 1)/2$. There is a family of n-vertex graphs $\mathcal{G}$ all of which have edge weights in $\{0, 1, \tau\}$ such that any quantum algorithm that for every graph $G \in \mathcal{G}$ computes with probability at least $2/3$ the weight of a minimum cut in G must make $\Omega(n^{3/2}\sqrt{\tau})$ queries in the adjacency matrix model. Similarly, any quantum algorithm that for every graph $G \in \mathcal{G}$ computes with probability at least $2/3$ the shores $(X, \overline{X})$ of a cut realizing the minimum weight must make $\Omega(n^{3/2}\sqrt{\tau})$ queries in the adjacency matrix model.*

Proof. Let V be an n-element set and partition V into disjoint sets $V = V_0 \sqcup V_1$ where $|V_0| = \lfloor n/2 \rfloor$, $|V_1| = \lceil n/2 \rceil$. Choose a distinguished vertex $v_0 \in V_0$, and let $V_0' = V_0 \setminus \{v_0\}$. Let $N = |V_0' \times V_1|$ and let $g : V_0' \times V_1 \to [N]$ be a bijection. For every $x \in \{0,1\}^N$ we define a weighted graph $G_x = (V, w_x)$ where

- $w_x(\{u, v\}) = \tau$ if $u, v \in V_0$ or $u, v \in V_1$,
- $w_x(\{u, v\}) = x(g(\{u, v\}))$ if $(u \in V_0', v \in V_1)$ or $(u \in V_1, v \in V_0')$,
- $w_x(\{u, v\}) = 0$ otherwise.

Let $k = \tau(\lfloor n/2 \rfloor - 1)$. In the following $\Delta_{G_x}(\cdot)$ will always be with respect to G_x and we drop the subscript. For any x it holds that $w_x(\Delta(V_0)) = |x|$ and $w_x(\Delta(\{v_0\})) = k$. Now consider any x and a subset $\emptyset \neq Y \subsetneq V$ different from V_0 or V_1. We can prove that $w_x(\Delta(Y)) \geq k$. To this end note that either $\emptyset \neq Y \cap V_0 \subsetneq V_0$ or $\emptyset \neq Y \cap V_1 \subsetneq V_1$. First assume that the former is the case. Then

$$w_x(\Delta(Y)) = \sum_{u \in Y, v \notin Y} w_x(\{u, v\}) \geq \sum_{u \in Y \cap V_0, v \in V_0 \setminus Y} w_x(\{u, v\}) \geq k,$$

as k is the weight of a minimum cut in the complete weighted graph over $\lfloor n/2 \rfloor$ nodes with all edge weights τ. If instead $\emptyset \neq Y \cap V_1 \subsetneq V_1$ then a similar argument shows that $w_x(\Delta(Y)) \geq \tau(\lceil n/2 \rceil - 1) \geq k$.

Thus if $|x| < k$ then $\Delta(V_0)$ will be the unique minimum cut of G_x, and the weight of a minimum cut in G_x will be $w_x(\Delta(V_0)) = |x|$. On the other hand, if $|x| > k$ then the weight of a minimum cut in G_x will be k, which is realized by the star cut $\Delta(\{v_0\})$ (and potentially other cuts in G_x) but not by $\Delta(V_0)$ as $w_x(\Delta(V_0)) = |x| > k$.

Let $\mathcal{S} = \{x \in \{0,1\}^N : |x| \in \{k-1, k+1\}\}$ and $\mathcal{G} = \{G_x : x \in \mathcal{S}\}$. Suppose there was a T query algorithm in the adjacency matrix model that for any $G_x \in \mathcal{G}$ with probability at least $2/3$ output the weight of a minimum cut in G_x. If the output is $< k$ then we know that $|x| = k-1$ and if the output is k then we know that $|x| = k+1$. Moreover, any query to the adjacency matrix of G_x can be simulated by a query to x, thus such an algorithm gives a T query algorithm to determine if $|x| = k-1$ or $|x| = k+1$ when we are promised one of these is the case. Since $\tau \leq (\lfloor n/2 \rfloor - 1)/2$ we have $2k+1 \leq N$ and therefore we may apply Theorem 34 to obtain $T \in \Omega(\sqrt{Nk}) = \Omega(n^{3/2}\sqrt{\tau})$.

Similarly, a T query algorithm in the adjacency matrix model that for any $G_x \in \mathcal{G}$ with probability at least $2/3$ outputs the shores of a cut realizing the minimum weight also implies a T query algorithm to determine if $|x| = k-1$ or $|x| = k+1$. In this case, if $|x| = k-1$ then the output must be $(V_0, \overline{V}_0)$ as these are the shores of the unique minimum cut in G_x. On the other hand, if $|x| = k+1$ then $(V_0, \overline{V}_0)$ is not a correct output. Thus the output of the algorithm lets us determine with probability at least $2/3$ if $|x| = k-1$ or $|x| = k+1$ and we again have $T \in \Omega(\sqrt{Nk}) = \Omega(n^{3/2}\sqrt{\tau})$. □

5.2 Adjacency array model

Given adjacency array access to a graph with edge-weight ratio τ, we showed an upper bound of $\tilde{O}(\sqrt{mn\tau})$ on the quantum query complexity of computing the weight of a minimum cut. In this section we prove two distinct lower bounds, each of which is tight in a specific regime. First we show that for any $\tau > 1$ there exists a family of dense graphs on n vertices with edge-weight ratio τ for which computing the weight of a minimum cut requires $\Omega(n^{3/2})$ queries to the adjacency array. This shows that the adjacency array upper bound of Theorem 21 of the source is tight for dense weighted graphs with constant (but non-unit) edge-weight ratio. Secondly and using a different approach, for any $1 \leq \tau \in O(n)$ we prove an $\Omega(\tau n)$ lower bound for a family of dense graphs with edge-weight ratio τ. This shows that we cannot get a quantum speedup when $\tau \in \Omega(n)$.

5.2.1 Constant edge-weight ratio

For the first bound we first need a claim about the minimum cuts of a complete weighted bipartite graph.

Claim 36. *Let $n \geq 8$ be a multiple of 4 and $G = (L \sqcup R, w)$ be a weighted bipartite graph with bipartition L, R where $|L| = 3n/4, |R| = n/4$. Further suppose that for every $x \in L, y \in R$ it holds that $w(\{x,y\}) \geq 1$. Then any cut of G that is not of the form $\Delta_G(\{x\})$ for $x \in L$ has weight at least $n/2$.*

Proof. First consider a star cut $\Delta_G(\{y\})$ for $y \in R$. This has weight at least $3n/4$, since this is the degree of y and all edges have weight at least 1.

It now remains to show the claim holds for non-star cuts. Consider a general non-star cut with shore $X \cup Y$ with $X \subseteq L, Y \subseteq R$. Let $k = |X|, \ell = |Y|$. As it is a non-star cut we have $k + \ell \geq 2$. By complementing as needed we may also assume that $k \leq 3n/8$. We also have the obvious constraints that $\ell \leq n/4$ and $k, \ell \geq 0$.

As G is a complete weighted bipartite graph with every edge weight at least one we have

$$w(\Delta_G(X \cup Y)) \geq k(n/4 - \ell) + \ell(3n/4 - k) \ .$$

As $k \leq 3n/8$ the term $\ell(3n/4 - k)$ is greater than $n/2$ whenever $\ell \geq 2$. Thus we can focus on $\ell \in \{0, 1\}$. If $\ell = 0$ then $k \geq 2$ and so the weight of the cut is at least $k(n/4) = n/2$ as desired. If $\ell = 1$ then the weight of the cut is $k(n/4 - 1) + 3n/4 - k$ which is always at least $3n/4$ as long as $n \geq 8$. □

This claim means that if $\min_{x \in L} w(\Delta_G(\{x\})) < n/2$ then this value will be the weight of a minimum cut in G. We can leverage this to show a lower bound as follows. In the next proof, for a function $f : \{0,1\}^n \to \{0,1\}$ we will use $Q_{1/3}(f)$ to denote the quantum query complexity of computing f with error at most $1/3$.

Theorem 37. *Let $n \geq 8$ be a multiple of 4 and $0 < \varepsilon \leq 1$. There is a family of n-vertex graphs $\mathcal{G}$ all of which have edge weights in $\{1, 1+\varepsilon\}$ such that any quantum algorithm that for every graph $G \in \mathcal{G}$ computes with probability at least $2/3$ the weight of a minimum cut in G must make $\Omega(n^{3/2})$ queries in the adjacency array model. Similarly, any quantum algorithm that for every graph $G \in \mathcal{G}$ computes with probability at least $2/3$ the shores $(X, \overline{X})$ of a cut realizing the minimum weight must make $\Omega(n^{3/2})$ queries in the adjacency array model.*

Proof. Let $X = \{x \in \{0,1\}^{n/4} : |x| = \lfloor n/8 \rfloor - 1\}$ and $Y = \{y \in \{0,1\}^{n/4} : |y| = \lfloor n/8 \rfloor + 1\}$. For every $x = (x^{(1)}, \ldots, x^{(3n/4)}) \in (X \cup Y)^{3n/4}$ we associate a bipartite graph $G_x = (L \sqcup R, w_x)$ where $L = \{1, \ldots, 3n/4\}, R = \{3n/4+1, \ldots, n\}$ and $w_x(\{i,j\}) = 1 + \varepsilon \cdot x^{(i)}(j - 3n/4)$ for every $i \in L, j \in R$. We set $\mathcal{G} = \{G_x : x \in (X \cup Y)^{3n/4}\}$.

Define the function $g : X \cup Y \to \{0,1\}$ where $g(x) = 0$ iff $x \in X$. We have $Q_{1/3}(g) \in \Omega(n)$ by Theorem 34. Let $f : \{0,1\}^{3n/4} \to \{0,1\}$ be the AND function, for which $Q_{1/3}(f) \in \Omega(\sqrt{n})$. By the composition theorem for quantum query complexity [HLŠ07, Rei11], we have $Q_{1/3}(h) \in \Omega(n^{3/2})$ for the composed function $h = f \circ g^{3n/4}$.

Let $x = (x^{(1)}, \ldots, x^{(3n/4)}) \in (X \cup Y)^{3n/4}$. If $h(x) = 1$ then $x^{(i)} \in Y$ for all $i \in [3n/4]$ and the weight of the star cut $\Delta_{G_x}(\{i\}) = n/4 + \varepsilon \cdot (\lfloor n/8 \rfloor + 1)$. As $\varepsilon \leq 1$ this will be the weight of a minimum cut in G_x by Claim 36. On the other hand if $h(x) = 0$ then some $x^{(i)} \in X$ and $\Delta_{G_x}(\{i\}) = n/4 + \varepsilon \cdot (\lfloor n/8 \rfloor - 1)$ and this will be the weight of a minimum cut of G_x. Thus computing the weight of a minimum cut of G_x lets us evaluate $h(x)$. Further, given oracle access to x we can simulate queries to G_x in the adjacency array model. Let A_x be a $3n/4$-by-$n/4$ matrix whose i^{th} row is the vector $1 + \varepsilon x^{(i)}$. Then the vertical concatenation of A with A^T is a valid adjacency array for G_x. To a degree query on vertex i we simply answer $n/4$ if $1 \leq 3n/4$ and $3n/4$ if $3n/4 + 1 \leq i \leq n$. We can also answer a query to the name and weight of the jth neighbor of i with one query to x. This shows that the $(1/3)$-error quantum query complexity of computing the weight of a minimum cut on graphs in $\mathcal{G}$ in the adjacency array model is at least $Q_{1/3}(f \circ g^{3n/4}) \in \Omega(n^{3/2})$.

Finally, suppose a quantum query algorithm can compute a shore of a minimum cut in G_x with T queries. We know that this shore must be of the form $\{v\}$ for a vertex $v \in L$. Thus with with $O(n)$ more queries the algorithm can classically compute the weight of a minimum cut by querying the weight of the

neighbors of v. Thus $T + O(n) \in \Omega(n^{3/2})$, which means $T \in \Omega(n^{3/2})$. This completes the proof. □

5.2.2 Large edge-weight ratio

Let $n \in \mathbb{N}$ be a multiple of 4 and V be a vertex set with $|V| = n$. Partition V into four sets V_1, V_2, V_3, V_4.

Now consider an integer τ such that $1 \leq \tau \leq 5n/8$ and $\tau n/10$ is an integer. Fix a set S of $\tau n/10$ "edge disjoint" quadruples $(u_1, u_2, u_3, u_4) \in V_1 \times V_2 \times V_3 \times V_4$. By edge disjoint we mean no pair of consecutive elements (u_i, u_{i+1}) or (u_4, u_1) appears in more than one quadruple. We fix an enumeration of S and refer to the vertices in the ℓth quadruple as $u_1^\ell, u_2^\ell, u_3^\ell, u_4^\ell$.

For every $x \in \{0,1\}^{\tau n/10}$ we define an n-vertex weighted graph $G_x = (V, w_x)$ where $w_x(\{u, v\}) = \tau$ if $u \neq v \in V_i$ for some $i \in [4]$, and for $\ell \in [\tau n/10]$ we set

$$w_x(\{u_1^\ell, u_2^\ell\}) = w_x(\{u_3^\ell, u_4^\ell\}) = x_\ell,$$
$$w_x(\{u_2^\ell, u_3^\ell\}) = w_x(\{u_4^\ell, u_1^\ell\}) = 1 - x_\ell \ .$$

Otherwise, $w_x(\{u, v\}) = 0$. In words, on each V_i we have a complete graph with all edge weights τ, and for each $\ell \in [\tau n/10]$ we either add unit weight edges $\{u_1^\ell, u_2^\ell\}, \{u_3^\ell, u_4^\ell\}$ or $\{u_2^\ell, u_3^\ell\}, \{u_4^\ell, u_1^\ell\}$ depending on x_ℓ. The construction is depicted in Fig. 1.

There are a few important points to note about this definition. First, the edge-weight ratio of G_x is τ. Second, for any X that nontrivially intersects some V_i we have that $w(\Delta_{G_x}(X)) \geq \tau(n/4 - 1)$. This means that such an X cannot be the shore of a minimum cut of G_x. Third, by construction the degree of every vertex of G_x is independent of x. This means that degree queries to G_x can be trivially answered and give us no information about x.

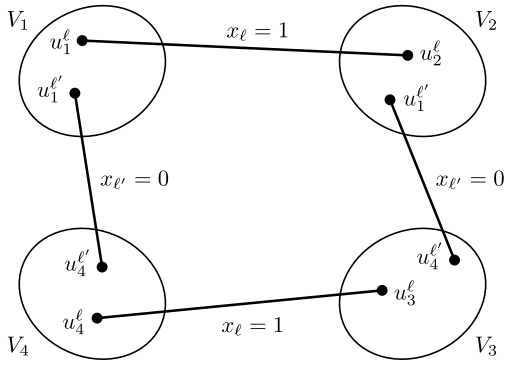


Figure 1: Figure of graph G_x. If $x_\ell = 1$ then we add edges $\{u_1^\ell, u_2^\ell\}$ and $\{u_3^\ell, u_4^\ell\}$. If $x_\ell = 0$ then we add edges $\{u_2^\ell, u_3^\ell\}$ and $\{u_4^\ell, u_1^\ell\}$.

Lemma 38. *We can simulate a single query to G_x in the adjacency array model using a single query to x.*

Proof. We first handle degree queries. This can be answered with no queries to x as the degree of a vertex is independent of x.

Now consider a query $(v, k) \in V \times [\deg(v)]$ to which we must answer the name u of the k-th neighbor of v and the edge weight $w_x(\{u, v\})$. For clarity of exposition, we assume $v = u_1^t \in V_1$; the other cases are handled similarly.

- If $k \leq n/4 - 1$ then return the k-th neighbor u of v inside V_1 and edge weight $w_x(\{u, v\}) = \tau$.
- If $k \geq n/4$ then let $j = k - n/4 + 1$. Letting ℓ denote the index of the jth quadruple of S containing v we query x_ℓ.
  - If $x_\ell = 1$ then return neighbor u_2^ℓ and edge weight $w(\{v, u_2^\ell\}) = 1$.
  - If $x_\ell = 0$ then return neighbor u_4^ℓ and edge weight $w(\{v, u_4^\ell\}) = 1$.

In total this takes a single query to x, which proves the lemma. □

Now we can prove the following lemma.

Lemma 39. *Fix integers n and τ such that $1 \leq \tau \leq 5n/8$ and $\tau n/10 \in \mathbb{N}$. Consider a string $x \in \{0,1\}^{\tau n/10}$ and the corresponding graph G_x. If $|x| < \tau n/20$ then G_x has a unique minimum cut with shores $(X, \overline{X}) = (V_1 \cup V_2, V_3 \cup V_4)$ and weight $w(\Delta_{G_x}(X)) = 2|x|$. If $|x| > \tau n/20$ then G_x has a unique minimum cut with shores $(X, \overline{X}) = (V_1 \cup V_4, V_2 \cup V_3)$ and weight $w(\Delta_{G_x}(X)) = 2(\tau n/10 - |x|)$.*

Proof. First consider any cut shore that nontrivially intersects some V_i. Since the subgraph $G_x[V_i]$ induced on V_i is a complete graph with edge weights τ, this implies that such a cut has weight at least $\tau(|V_i| - 1) = \tau(n/4 - 1)$. Now consider the small set of remaining cut shores that trivially intersect the V_i's. The weight of each one of these cuts can be easily expressed as a function of the Hamming weight $|x|$ of the input:

$$\begin{aligned} w(\Delta_{G_x}(V_i)) &= \tau n/10, \\ w(\Delta_{G_x}(V_1 \cup V_3)) &= w(\Delta_{G_x}(V_2 \cup V_4)) = 2\tau n/10, \\ w(\Delta_{G_x}(V_1 \cup V_2)) &= w(\Delta_{G_x}(V_3 \cup V_4)) = 2|x|, \\ w(\Delta_{G_x}(V_1 \cup V_4)) &= w(\Delta_{G_x}(V_2 \cup V_3)) = 2(\tau n/10 - |x|). \end{aligned}$$

It is clear that all minimum weight cuts will be among these cuts, and the lemma easily follows. □

Using this lemma we can prove the following theorem.

Theorem 40. *Let $\tau, n \in \mathbb{N}$ be such that $1 \leq \tau \leq 5n/8$ and $\tau n/20 \in \mathbb{N}$. There exists a family of n-vertex graphs $\mathcal{G}'$ with $\Omega(n^2)$ edges, all of which have edge weights in $\{1, \tau\}$, such that any quantum algorithm that for every graph $G' \in \mathcal{G}'$ computes with probability at least $2/3$ the weight of a minimum cut in G' must make $\Omega(n\tau)$ queries in the adjacency array model. Similarly, any quantum algorithm that for every graph $G' \in \mathcal{G}'$ computes with probability at least $2/3$ the shores $(X, \overline{X})$ of a cut realizing the minimum weight must make $\Omega(n\tau)$ queries in the adjacency array model.*

Proof. First consider the set of strings $\mathcal{X} \subseteq \{0,1\}^{\tau n/10}$ with Hamming weight

$$|x| = \lfloor \tau n/100 \rceil \pm 1 < \tau n/20.$$

By Lemma 39 the graph G_x, $x \in \mathcal{X}$, has a unique minimum cut with shores $(V_1 \cup V_2, V_3 \cup V_4)$ and weight $2|x|$. Now let $\mathcal{G}' = \{G_x : x \in \mathcal{X}\}$ and assume the existence of a quantum algorithm that for every $G_x \in \mathcal{G}'$ computes with probability at least $2/3$ the weight $2|x|$ of a minimum cut in G' with at most q queries to the adjacency array of G_x. By Lemma 38 this is equivalent to outputting the Hamming weight $|x|$ with probability at least $2/3$ for any $x \in \mathcal{X}$ while making only q queries to x. Using Theorem 34 this implies the lower bound $q \in \Omega(\tau n)$.

Next consider the set of strings $\mathcal{X}' \subseteq \{0,1\}^{\tau n/10}$ that have Hamming weight $|x| = \tau n/20 \pm 1$. By Lemma 39 the graph G_x, $x \in \mathcal{X}'$, again has a unique minimum cut. If $|x| = \tau n/20 - 1$ then its shores are $(V_1 \cup V_2, V_3 \cup V_4)$, while if $|x| = \tau n/20 + 1$ then its shores are $(V_1 \cup V_4, V_2 \cup V_3)$. Now assume that there exists a quantum algorithm that with probability at least $2/3$ returns the shores of a minimum weight cut of G_x with at most q queries to the adjacency array of G_x. By Lemma 38 this is equivalent to distinguishing $|x| = \tau n/20 - 1$ from $|x| = \tau n/20 + 1$ with probability at least $2/3$ for any $x \in \mathcal{X}$ while making only q queries to x. Using Theorem 34 this implies the lower bound $q \in \Omega(\tau n)$. □

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